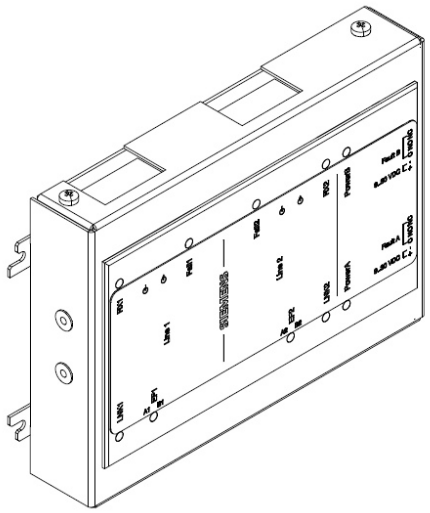
SIEMENS



FN2013-U1, FN2014-UL Fiber switch module (SM/MM)

Installation

Mounting

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2024-11-11

Legal notice

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1 Description

Mounting Installation FN2013-U1, FN2014-U1 Fiber Network Module (SM/MM)

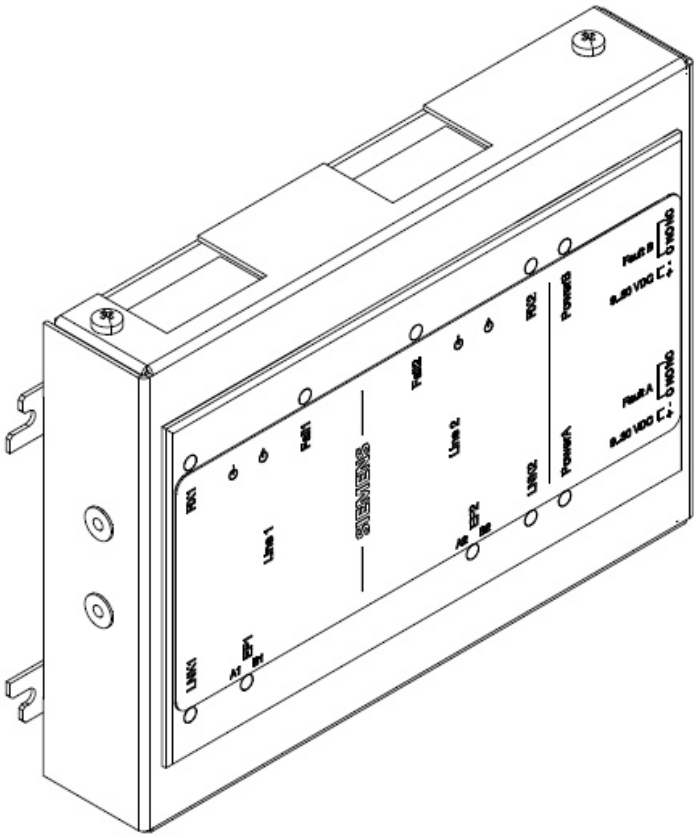


Fig. 1: Fiber Network Module FN2013/FN2014

The fiber network module multi-mode (MM) and the fiber network module single-mode (SM) can be used for the exclusive purpose of networking fire detection panels to the HNET, XNET, SNC, or HUB-4 over large distances using fiber-optic cable.

The fiber network modules have the following features:

- Fiber network module (SM) FN2013: Single-mode transmission up to 131,000 ft/ 40,000 m
- Fiber network module (MM) FN2014: Multi-mode transmission up to 13,100 ft/ 4,000 m
- SC connections for fiber-optic cable
- Trouble signaling via LED
- Mounting in panel

2 Installation in Desigo/Cerberus PRO Modular

In Desigo/Cerberus PRO Modular systems, the fiber network module is installed on a COM-BRK bracket that can be mounted to the CAB-1, CAB-2 or CAB-3 back box or to the optional CAB-MP.

The COM-BRK takes up the same space as a CC-2 card cage. The fiber network module should first be secured to the COM-BRK using the three screws provided with the module, then installed in the back box or mounting plate using the four screws provided with the bracket.

Once the bracket and module are secured to the cabinet, wire the module and connect the optical fiber cable as described later in this document.

NOTICE	
!	The fiber network module and bracket cannot be installed behind a FCM2041-U2, FCM2041-U3 or FMT.

NOTICE	
!	Only use the fiber network module in dry rooms.

NOTICE	
!	The product(s) have been tested at relative humidity 85 ±5 percent and a temperature of 30 ±2°C (86 ±3°F).

NOTICE	
!	The unit shall be operated at ambient temperatures between 0 and 49° C (32 and 120° F).

⚠ CAUTION	
⚠	When installing, route field wiring away from sharp projections, corners, and internal components.

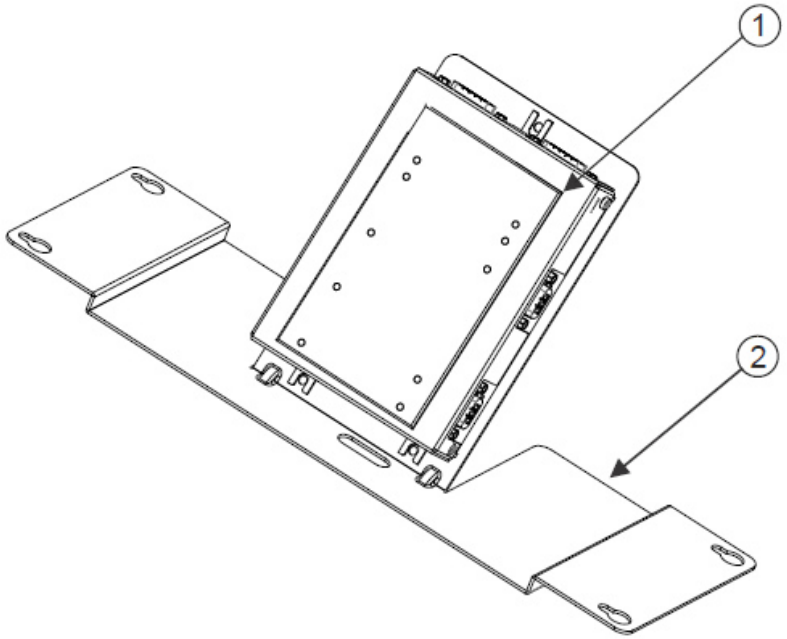


Fig. 2: Mounting of fiber network module with COM-BRK in housing CAB-1/-2/-3

1 - Fiber network module
2 - COM-BRK mounting plate

3 Views

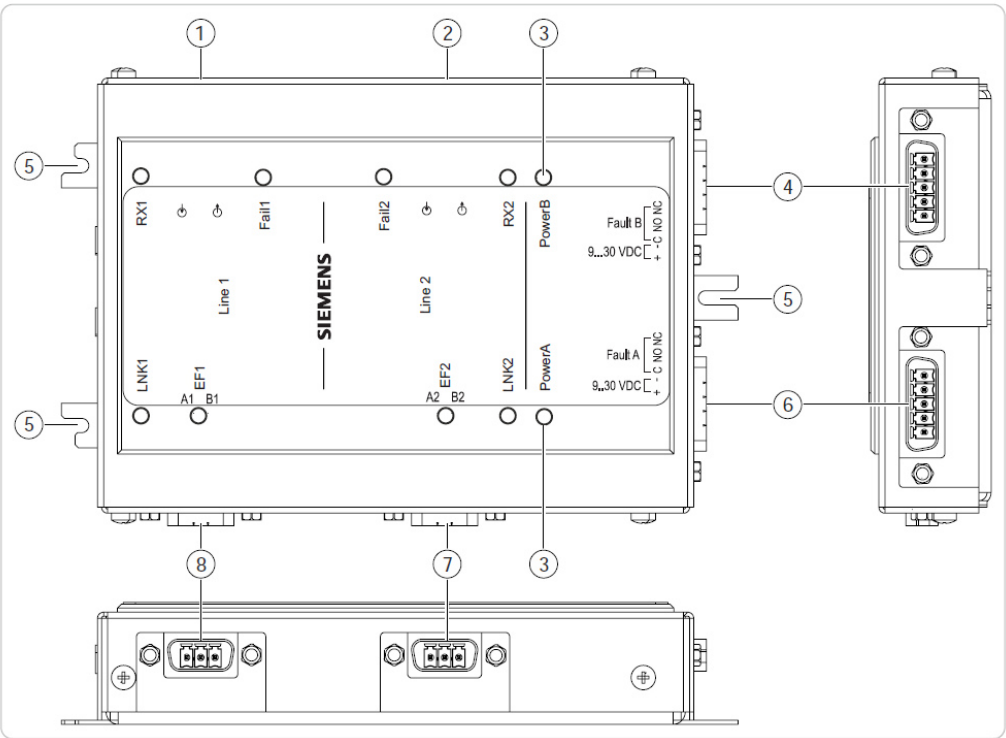


Fig. 3: Views of the fiber network module from above and the side

1 - Fiber-optic SC connection for circuit 1
2 - Fiber-optic SC connection for circuit 2
3 - 10 LED indicators for data transmission, errors and power supply for both channels
4 - Power B socket, do not use
5 - Fastening tab for housing mounting
6 - Power A socket, power supply
7 - A2/B2 socket for HNET, XNET, HUB-4, SNC
8 - A1/B1 socket for HNET, XNET, HUB-4, SNC

4 Wiring

4.1 Wiring for Desigo/Cerberus PRO Modular

NOTICE	
!	Only connect to a UL864/ULC-S527 approved, power-limited, and regulated power unit, e.g., PSC-12 / PSC-12M (TB3).

NOTICE	
!	Wiring methods used shall be in accordance with the National Electrical Code, ANSI/ NFPA 70, the Standard for Installation and Classification of Burglar and Holdup Alarm Systems, UL 681, and the Standard for Central-Station Alarm Services, UL 827, as applicable.

- In Desigo/Cerberus PRO Modular systems, 'Power B' connections to the fiber module and monitoring of its trouble contacts are not required, as the fiber modules must be installed in the same enclosure as the power supplies that provide power to them.
- Power A' inputs can only be supplied by UL864/ULC-S527 approved regulated power sources.
  - PSC-12 / PSC-12M (TB3) or PSX-12/ PSX-12M (TB3)
  - UL/ULC LISTED REGULATED DC 24 V POWER SUPPLY (POWER LIMITED)

To ensure compatibility with the NIC-C, the HUB-4, or the SNC, connect the polarity of circuit exactly as shown.

The fiber network module FN2013/FN2014 does not feature ground fault supervision. For connections via HNET/XNET, ground fault supervision is via the NIC-C.

Use only EF1 or EF2. EF1 and EF2 are interconnected internally. EF1 has an internal 120Ω terminating resistor; EF2 does not have a terminating resistor. The right selection of EF1 or EF2 depends on the application. You will find information about selecting EF1 or EF2 in the following graphics in this selection and in the graphics in selection 'Appendix'.

FPGA Revision	Relay transfers on power loss only	Relay transfers on power loss or RX fiber break.
No Marking	Yes	No
Rev 2.1.4 or higher *	Yes	Yes

NOTICE	
!	All terminals are power limited.

*The relay contact on the Power A socket will transfer to the off-normal; state when the single fiber cable connected to "Receive IN SC plug" of the RX1 Line 1 is broken or disconnected.

*The relay contact on the Power B socket will transfer to the off-normal state when the single fiber cable connected to "receive IN SC plug" of the RX2 Line 2 is broken or disconnected.

These relays do not transfer when the single fiber cable on the Transmit Out SC plug" TX1 or TX2 is broken or disconnected. This fault condition is detected at the receiving units RX1/RX2 input.

In applications where both Line 1 and Line 2 are used, both power inputs must be supplied power to maintain the relays in their normal state.

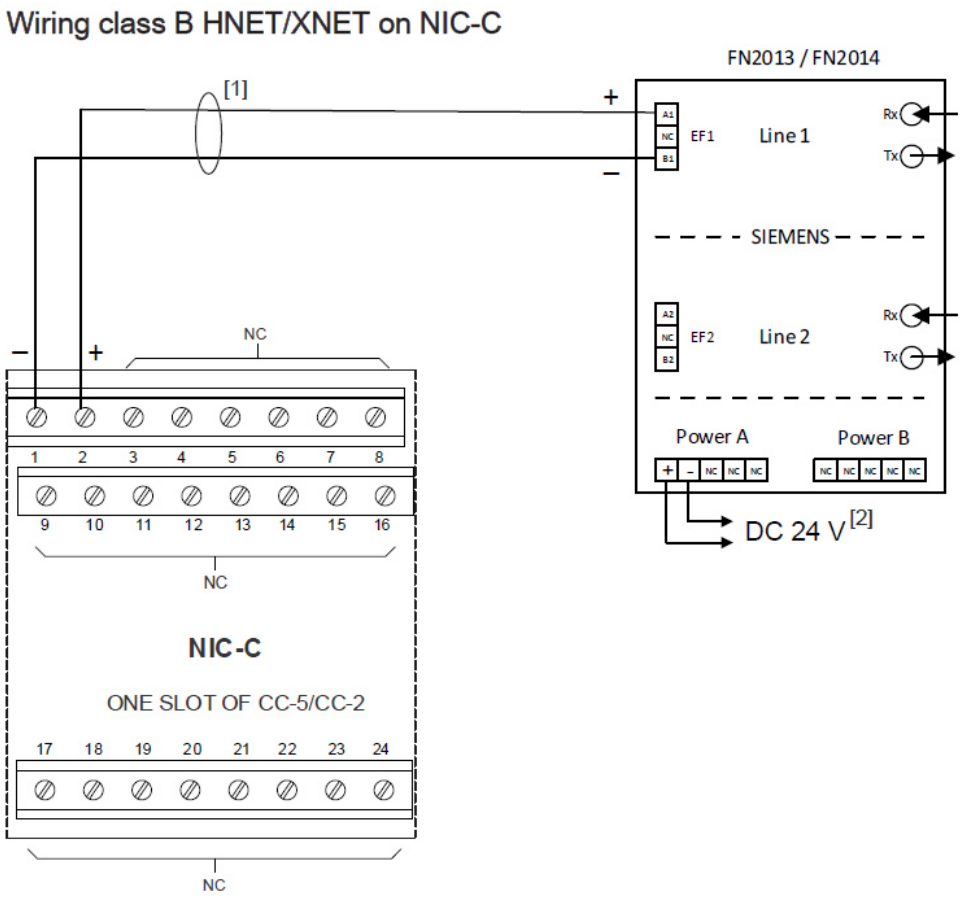


Fig. 4: Class B, HNET/XNET wiring on NIC-C

Pin	Designation	Description
NIC-C/1	-	NIC-C, RS48 circuit 1 -, monitored by NIC-C[1]
NIC-C/2	+	NIC-C, RS48 circuit 1 +, monitored by NIC-C[1]
NIC-C/3-24	NC	NIC-C, do not use
EF1/A1	+	Fiber Network module, RS485 circuit 1 +

EF1/B1	-	Fiber Network module, RS485 circuit 1 -
Power A/+	+	Fiber Network module, + DC 24 V[2]
Power A/-	-	Fiber Network module, GND [2]
Power B	NC	Fiber Network nodule, do not use

[1] RS485 ratings:

- 8 V P-P, 150 mA max
- 2000 ft/610 m max
- 16 AWG max, 18AWG min
- Positive or negative ground fault supervision at < 10 kΩ

[2] Only connect to a UL864/ULC-S527 approved, power-limited, and regulated power unit, e.g., PSC-12 / PSC-12M (TB3). No EOL resistor required on NIC-C or fiber network module.

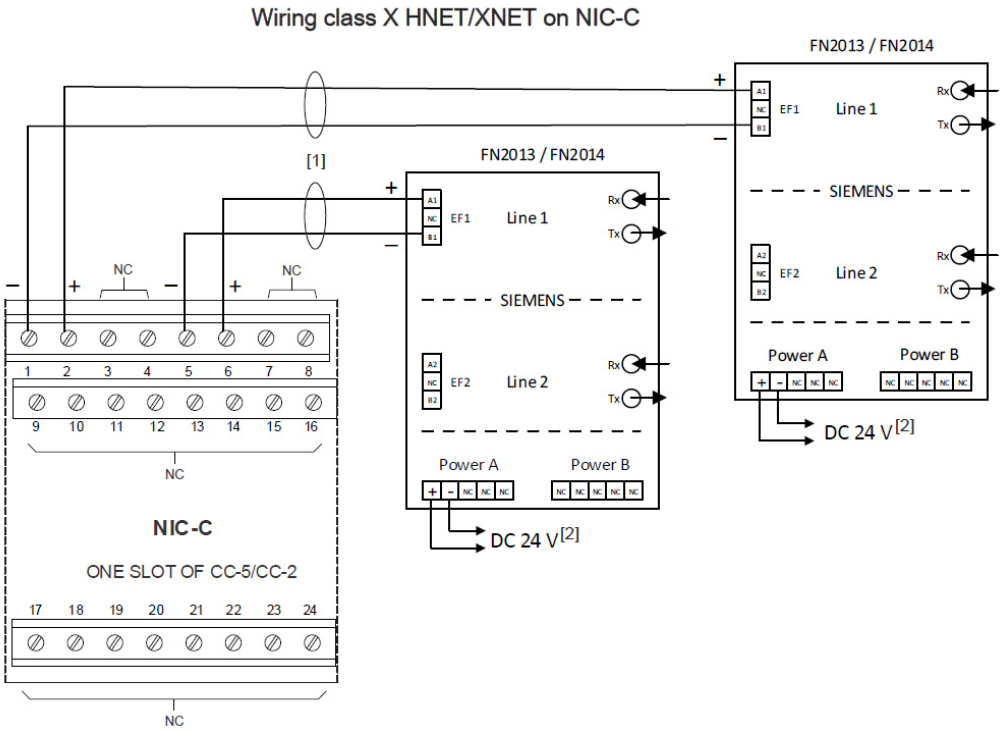


Fig. 5: Class X, HNET/XNET wiring on NIC-C

Pin	Designation	Description
NIC-C/1	-	NIC-C, circuit 1A-, monitored by NIC-C[1]
NIC-C/2	+	NIC-C, circuit 1A+, monitored by NIC-C[1]
NIC-C/3-4	NC	NIC-C, do not use
NIC-C/5	-	NIC-C, circuit 1B-, monitored by NIC-C[1]

NIC-C/6	+	NIC-C, circuit 1B+, monitored by NIC-C[1]
EF1/A1	+	Fiber Network module, circuit 1A +/circuit 1B +
EF1/B1	-	Fiber Network module, circuit 1A -/circuit 1B -
Power A/+	+	Fiber Network module, + DC 24 V[2]
Power A/-	-	Fiber Network module, GND[2]
Power B	NC	Fiber Network module, do not use

[1] RS485 ratings:

- 8 V P-P, 150 mA max
- 2000 ft/610 m max
- 16 AWG max, 18AWG min
- Positive or negative ground fault supervision at < 10 kΩ

[2] Only connect to a UL864/ULC-S527 approved, power-limited, and regulated power unit, e.g., PSC-12 / PSC-12M (TB3). No EOL resistor required on NIC-C or fiber network module.

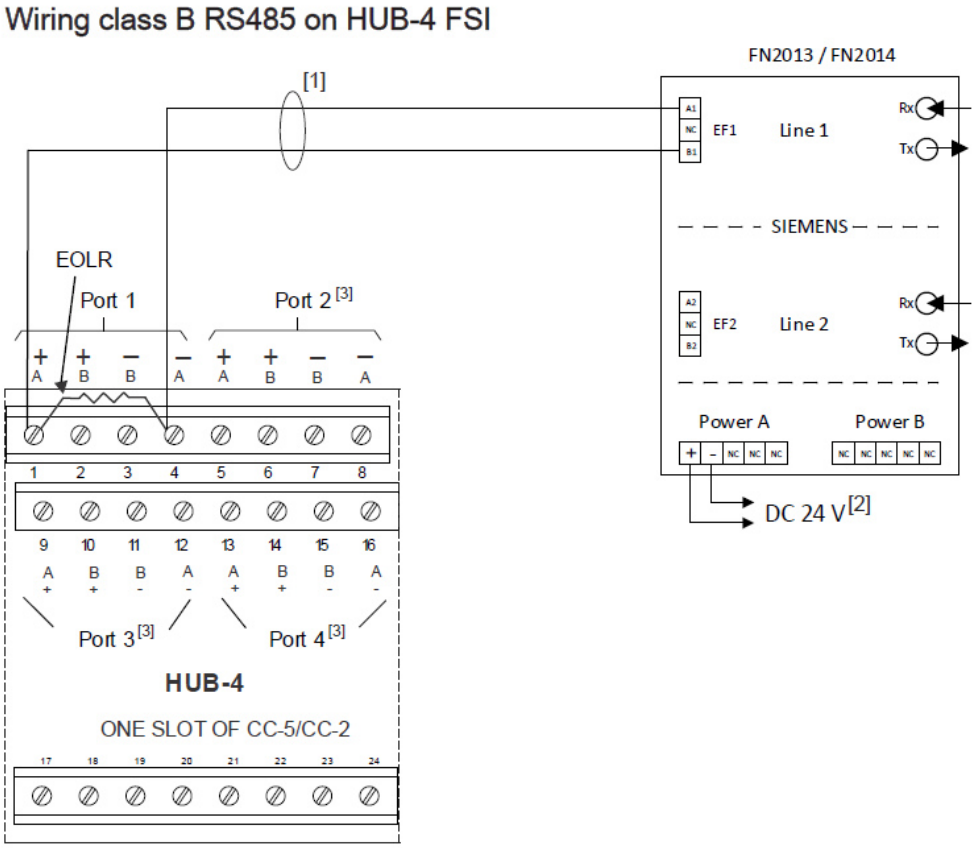


Fig. 6: Class B, RS485/HUB-4 FSI wiring

Pin	Designation	Description

Port 1/1	A+	HUB-4, circuit 1A +. EOLR, not monitored [1]
Port 1/2	B+	Not connected
Port 1/3	B-	Not connected
Port 1/4	A-	HUB-4, circuit 1A -. EOLR, not monitored [1]
EF1/A1	+	Fiber network module, circuit 1A +, not monitored
EF1/B1	-	Fiber network module, circuit 1A-, not monitored
Power A/+	+	Fiber network module, + DC 24 V [2]
Power A/-	-	Fiber network module GND [2]
Power B	NC	Fiber network module, do not use

[1] RS485 ratings:

- 8 V P-P, 150 mA max.,
- 20 ft/6m max. in conduit, not monitored
- 16 AWG max., 18AWG min.

[2] Only connect to a UL864/ULC-S527 approved, power-limited, and regulated power unit, e.g., PSC-12 / PSC-12M (TB3).

[3] Port x*: Modem or RS485 interface. The same interface type must be used on each COM-2 card. If only one interface card is being used on a COM-2 card, it must be inserted into port 1 pr port 3. EOLR 120Ω, 1/4 W, P/N 104-820150

Wiring class X RS485 on HUB-4 FSI

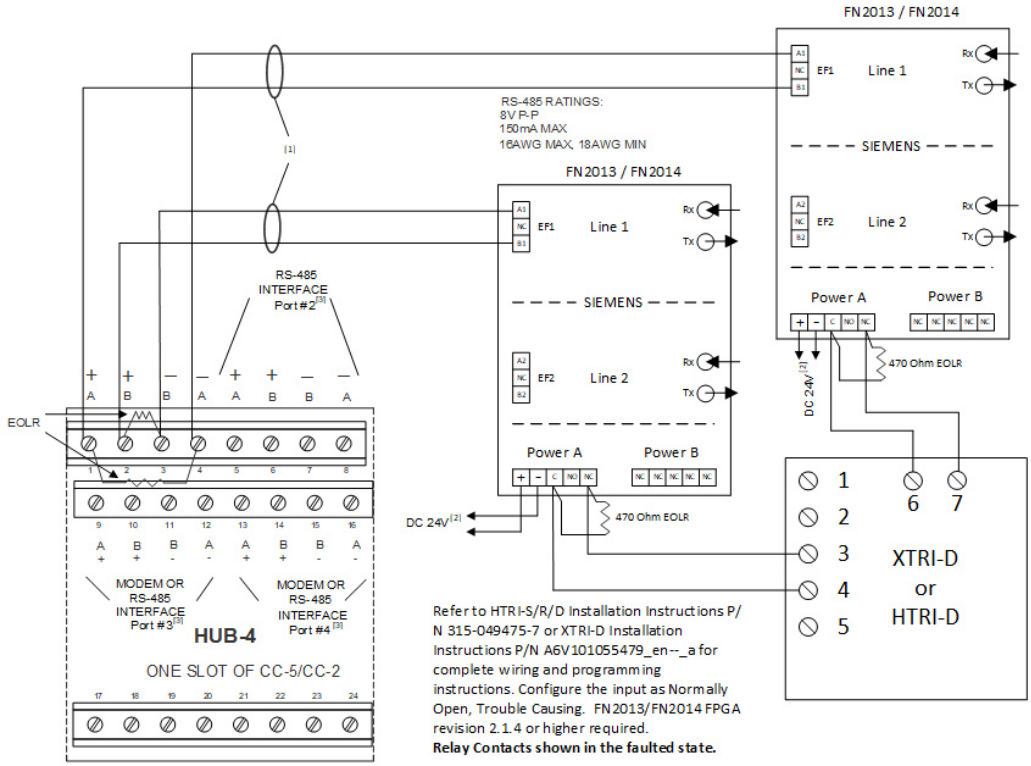


Fig. 7: Class X, RS485/HUB-4 FSI wiring (FN2013/FN2014 required)

Pin	Designation	Description
Port 1/1	A+	HUB-4, circuit 1A +. EOLR1, not monitored [1]
Port 1/2	B+	HUB-4, circuit 1B +. EOLR2, not monitored [1]
Port 1/3	B-	HUB-4, circuit 1B-, EOLR2, not monitored [1]
Port 1/4	A-	HUB-4, circuit 1A -, EOLR1, not monitored [1]
EF1/A1	+	Fiber network module, circuit 1A +/circuit 1B +, not monitored
EF1/B1	-	Fiber network module, circuit 1A-/circuit 1B -, not monitored
Power A/+	+	Fiber network module, + DC 24 V [2]
Power A/-	-	Fiber network module, GND [2]
Power B	NC	Fiber network module, do not use

[1] RS485 ratings:
-8 V P-P, 150 mA max

-20 ft/6 m max. in conduit, not monitored

16 AWG max., 18 AWG min

[2]Only connect to a UL864/ULC-S527 approved, power-limited and regulated power unit, e.g. PSC-12 / PSC-12M (TB3)

[3]Port x*: modem or RS485 interface. The same interface type must be used on COM-2 card. If only one interface card is being used on a COM-2 card, it must be inserted into port 1 or port 3. EOLR 120Ω , 1/4 W, P/N 104-820150

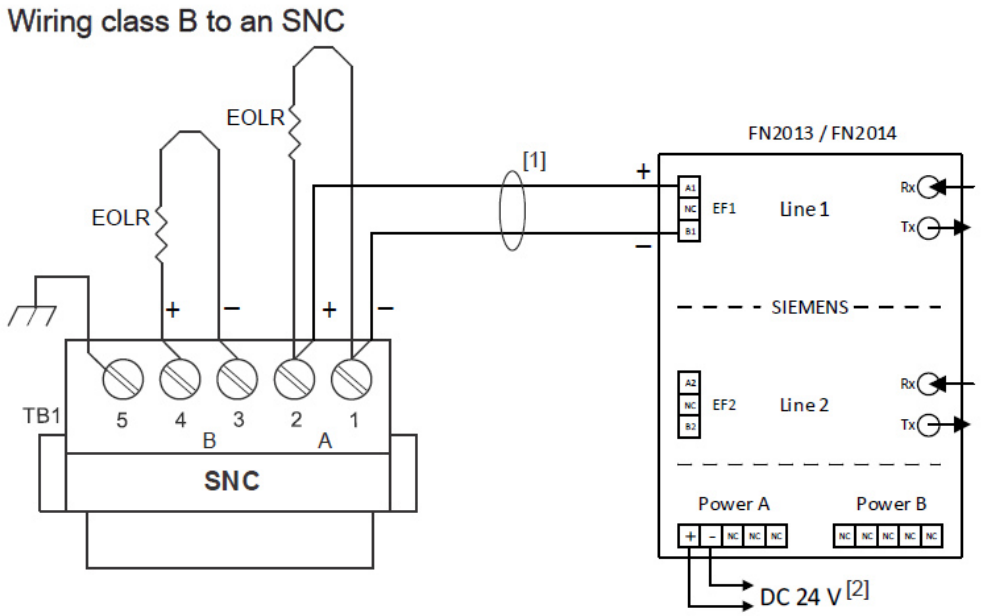


Fig. 8: Class B wiring to an SNC (Desigo CC/Cerberus DMS/NCC)

Pin	Designation	Description
SNC/1	-	SNC, circuit A -, EOLR, not monitored [1]
SNC/2	+	SNC, circuit A +, EOLR, not monitored [1]
SNC/3	-	SNC, circuit A -, EOLR
SNC/4	+	SNC, circuit A +, EOLR
SNC/5		Ground
EF1/A1	+	Fiber Network module, circuit A+, not monitored
EF1/B1	-	Fiber Network module, circuit A-, not monitored
Power A/+	+	Fiber Network module, +DC 24 V [2]
Power A/-	-	Fiber network module, GND [2]
Power B	NC	Fiber network module, do not use

[1] RS485 ratings:

-8 V P-P, 150 mA max

-20 ft/6 m max. in conduit, not monitored

16 AWG max., 18 AWG min

[2]Only connect to a UL864/ULC-S527 approved, power-limited and regulated power unit, e.g. PSC-12 / PSC-12M (TB3) EOLR 120Ω. 1/2 W, P/N 104-820350

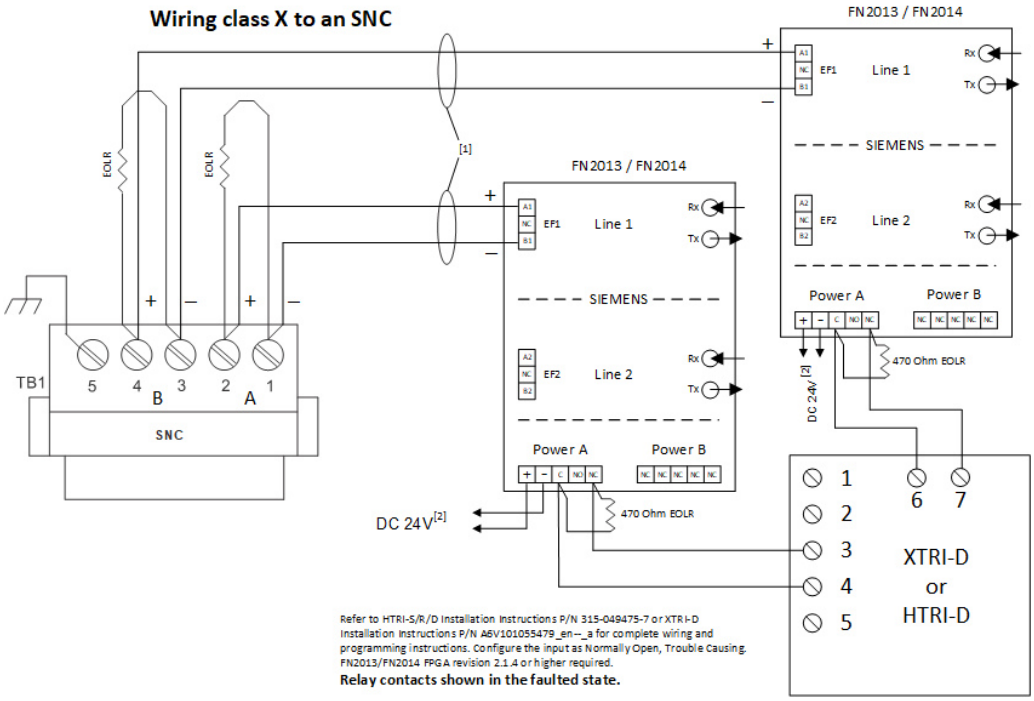


Fig. 9: Class X wiring to an SNC (Desigo CC/Cerberus DMS/NCC) (two FN2013/FN2014 required)

Pin	Designation	Description
SNC/1	-	SNC, circuit A-, EOLR, not monitored [1]
SNC/2	+	SNC, circuit A+, EOLR, not monitored [1]
SNC/3	-	SNC, circuit B-, EOLR, not monitored [1]
SNC/4	+	SNC circuit B +, EOLR, not monitored[1]
SNC/5		Ground
EF1/A1	+	Fiber network module, circuit A +/circuit B +, not monitored
EF1/B1	-	Fiber network module, circuit A +/circuit B +, not monitored
Power A/+	+	Fiber network module, +DC 24 V [2]
Power A/-	-	Fiber network module, GND [2]

Power B	NC	Fiber network module, do not use

[1] RS485 ratings:

-8 V P-P, 150 mA max

-20 ft/6 m max. in conduit, not monitored

16 AWG max., 18 AWG min

[2]Only connect to a UL864/ULC-S527 approved, power-limited and regulated power unit, e.g. PSC-12 / PSC-12M (TB3) EOLR 120Ω. 1/2 W, P/N 104-820350

Fiber optic cable wiring

- The accessible laser radiation is harmless. It corresponds to laser class 1 in accordance with IEC 60825/ANSI Z136.
- The fiber-optic cable is wired cross-wise:
  - The optical transmitter (TX) of one device must always be connected with the optical receiver (RX) of the other device.
- The bending radius of the fiber-optic cable must not be less than that specified by the manufacturer (e.g. 10 x outer diameter).

4.2 Power supply sockets (4,6)

The 5-pole connectors for the power supply are supplied with the fiber network module.

Pin	Designation	Description
1	DC 24 V+	Supply input + of DC 24 V AUX from periphery board or CC-2 / CC-5
2	DC 24 V-	Supply input - of DC 24 V AUX from periphery board or CC-2 / CC-5
3	COM	Common contact
4	OPEN	Closes on Rx / Power Fault
5	CLOSED	Opens on Rx / Power Fault

Admissible cable cross-section: 16...18 AWG

- When the module is installed in fire control panels or fire voice control panels alone, the power supply does not have to be supervised or routed redundantly.
- The supply line for a remote network module must be shielded.

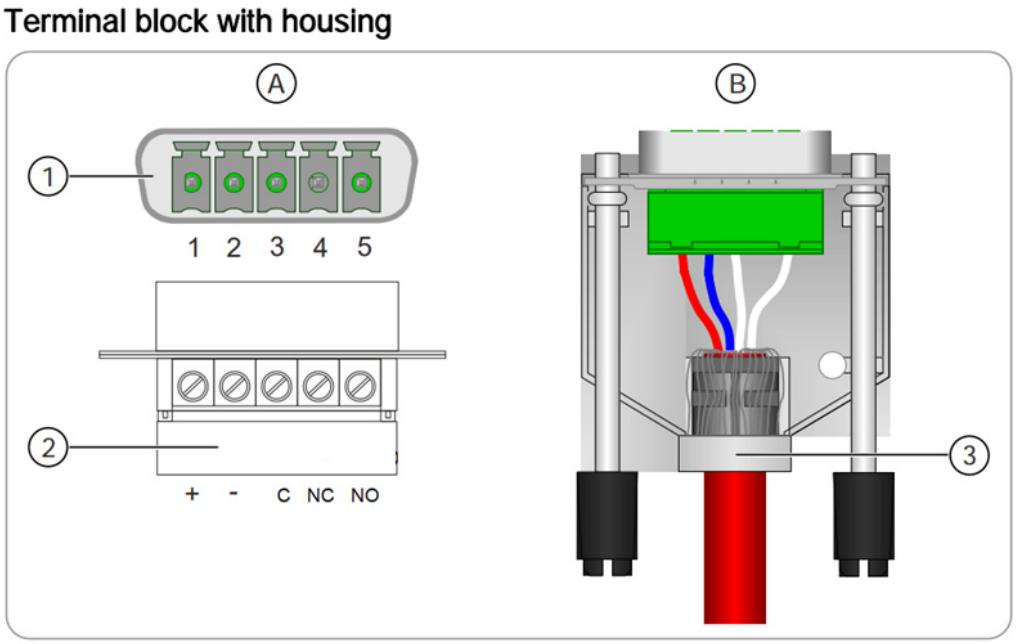


Fig. 10: Connector power A (B)

A - Connector plug without housing
B - Connector plug with housing
1 - Connector
2 - Cover cap, open
3 - 360° shielding
Note: Do not use housing when installing on COM-BRK.

4.3 Sockets for HNET, XNET, HUB-4, SNC (7,8)

The 3-pole connectors for the HNET/XNET/HUB-4/SNC connection are supplied with the fiber network module.

The total cable length of the electrical HNET/XNET/HUB-4/SNC connections (copper cable) in the optical network must not exceed 2,000 ft/610 m

Connection A1/B1 has an internal 120 Ω terminating resistor; connection A2/B2 does not have a terminating resistor. Connection A1/B1 is for use at a terminated endpoint; connection A2/B2 is for a cascade (daisy chain).

Pin	Designation	Description
3	B1	HNET/XNET/HUB-4/SNC circuit 1 (-)
2	NC	Do not use
1	A1	HNET/XNET/HUB-4/SNC circuit 1 (+)

Table 1: Connection A1/B1(8)

Admissible cable cross-section: 16…18 AWG

Pin	Designation	Description
3	B2	HNET/XNET/HUB-4/SNC circuit 2 (-)
2	NC	Do not use
1	A2	HNET/XNET/HUB-4/SNC circuit 2 (+)

Table 2: Connection A2/B2 (7)

Admissible cable cross-section: 16…18 AWG

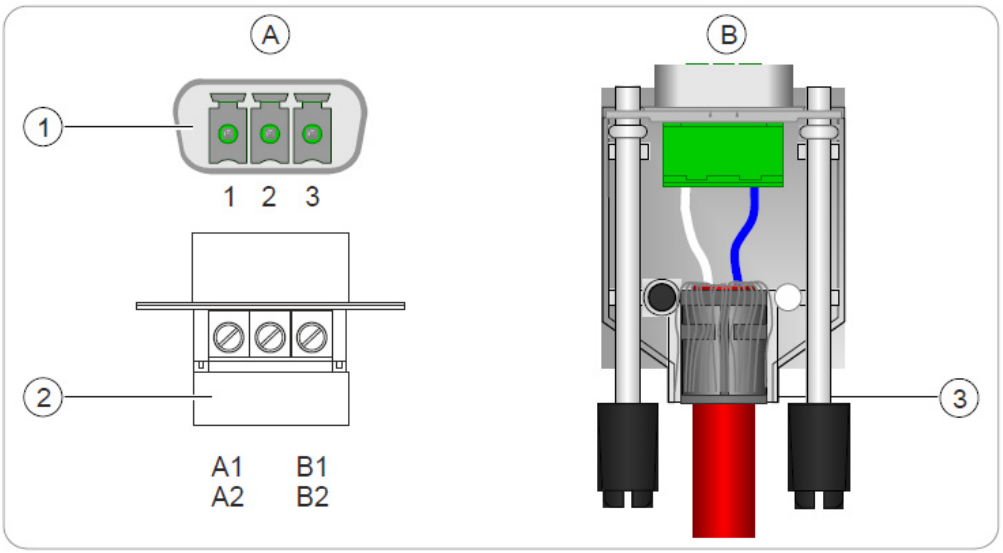


Fig. 11: Connector EF1/EF2, FCnet/C-WEB A1/B1 (A2/B2)

A - Connector plug without housing
1 - Connector
2 - Cover cap, open
B - Connector plug with housing
3 - 360° shielding

5 Indicators

LED	Color	Function	Condition	Meaning
RX1	Green	Receipt of data from fiber optic circuit '1'	Lit up	Data is being received from circuit '1'
Fail 1	Yellow	Fault on fiber optic circuit '1'	Lit up	Weakly receiving data from circuit '1'
Fail 2	Yellow	Fault on fiber optic circuit '2'	Lit up	Weakly receiving data from circuit '2'
RX2	Green	Receiving data from fiber-optic circuit '2'	Lit up	Data is being received from circuit '2'
Power B	Green	Supervision of feeding voltage for 'Power B'	On Off	Normal operation (feeding voltage available) No feeding voltage or voltage below minimum input voltage (9V)
Power A	Green	Supervision of feeding voltage for 'Power A'	On Off	Normal operation (feeding voltage available) No feeding voltage or voltage below minimum input voltage (9V)/
LNK1	Green	Data transmission HNET/XNET/ HUB-4/SNC - circuit '1'	Flashes	Data is being transmitted via A1/B1
EF1	Yellow		Off	Not used
EF2	Yellow		Off	Not used
LNK2	Green	Data transmission HNET/XNET/ HUB-4/SNC - circuit '2'	Flashes	Data is being transmitted via A2/B2

6 Technical Data

Supply input	Input voltage	DC 24V
	Decentralized mounting	Shield supply cable (not in scope of supply)
	Operating current	Max. 132 mA @ DC 24 V
Supervision contact	Power/ Rx Fault	DC 30V max DC 24 max.
Fiber optic cable	Type of connection	SC connector
	Cable end box	Standard ‘PC’ or 8° polish ‘APC’
	Wavelength	1310 nm (for MM and SM)
FN2014-U1	**Multi-mode**	
	Range:	
	Fiber type 62.5/125µm	13100 ft/4000 m
	Fiber type 50/125µm	6550 ft/2000m
	Optical budget:	
	Fiber type 62.5/125µm	11 dBm
	Fiber type 50/125µm	7.5 dBm
FN2013-U1	**Single mode:**	
	Range:	
	Fiber type 9/125µm	131000 ft/4000m
	Optical budget:	
	Fiber type 9/125µm	29dBm
	Minimum optical attenuation (corresponds to a minimum fiber length	3 dBm
RS485 connection [EF1],]EF2] (HNET, XNET, HUB-4, SNC)	Length of line: HNET XNET (NIC-C)	2000 ft/610m at19.2 kbit/s
	Length of line: HUB-4, SNC	20 ft/610m in conduit
	Impedance	60Ω
	Transmission mode	Half-duplex
	Voltage/current	8 V p-p, 15mA
	Superivision[1]	No ground fault supervision
	Terminating resistor	No EOL required
	Cable type	Shielded and unshielded cables

[1] For connections via HNET/XNET, ground fault supervision is via NIC-C.

7 Appendix A

Applications for FN2013 and FN2014 Modules

The diagrams below show typical applications for the FN2013 and FN2014. Class B networks are shown; for class X, two of every device are required and must be connected in a parallel (redundant) network. Fiber-optic cables L1 and L2 are interchangeable. L1 does not always have to be connected to L2. All fiber-optic cables in the diagrams below are connected with a fiber pair between the FN2013/14 modules.

Mixing of single-mode and multi-mode fibers is not permitted

Connect the EF1 port on FN2013 or FN2014. An external terminating resistor is not necessary because there is an internal 120 Ω terminating resistor installed at the EF1 port.

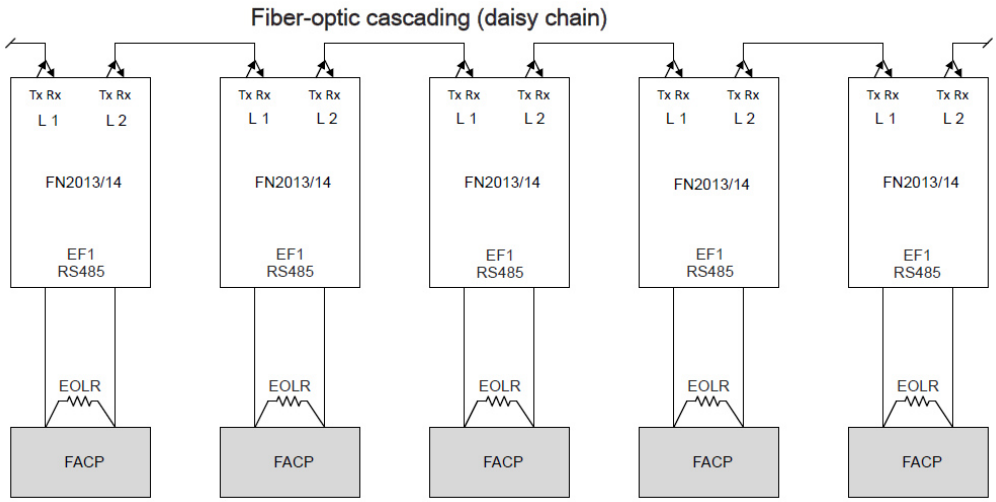


Fig. 12: Fiber-optic cascading (daisy chain)

L1 Fiber optic SC connection for circuit 1
L2 Fiber optic SC connection for circuit 2
EF1 Connection A1/B1, RS485
FACP Fire control panel
EOLR Terminating resistor 120Ω

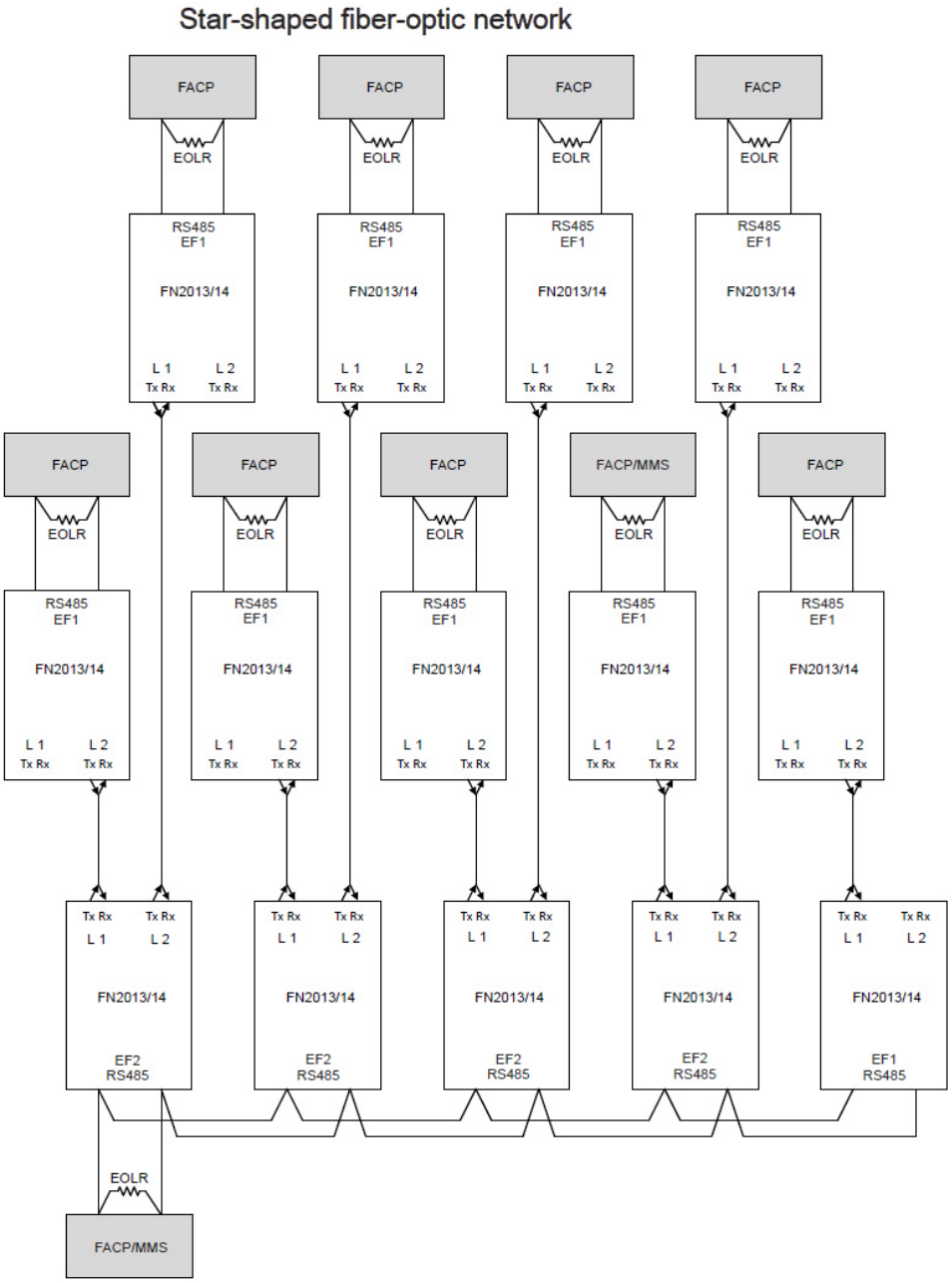


Fig. 13: Star-shaped fiber-optic network

L1 Fiber optic SC connection for circuit 1

L2 Fiber optic SC connection for circuit 2

EF1 Connection A1/B1, RS485

EF2 Connection A2/B2, RS485

FACP Fire control panel

MMS Management station

EOLR Terminating resistor 120Ω

All modules connected via fiber-optic cables must be the same model (FN2013 or FN2014) and use the same type of fiber (multi-mode or single-mode). Mixing of module types and types of fiber is permitted on the RS485 network.

FN2013 or FN2014 modules in the center of the RS485 daisy chain must be connected to the EF2 port. If an FN2013 or FN2014 is located at one end of the chain, connect it to the EF1 port which has an internal 120-ohm terminating resistor; an external terminating resistor is not necessary.

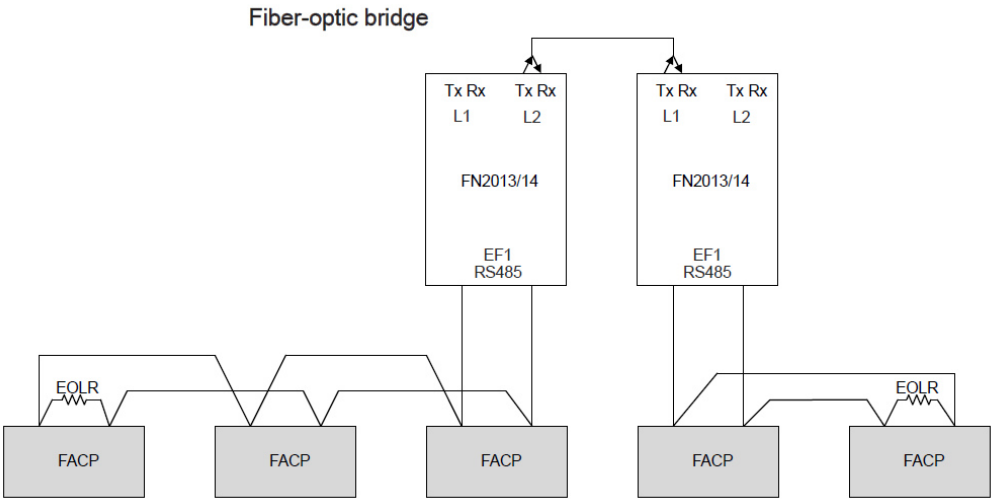


Fig. 14: Fiber-optic bridge

L1 Fiber optic SC connection for circuit 1

L2 Fiber optic SC connection for circuit 2

EF1 Connection A1/B1, RS485

FACP Fire control panel

EOLR Terminating resistor 120Ω

Integrating FN2013/FN2014 into existing D23xx networks

The diagrams below show typical applications for D2300CPS/D2325CPS modules and how FN2013 or FN2014 modules can be added to existing fiber-optic networks with D23XXCPS modules. Class B networks are shown; for class X, two of every device are required and must be connected in a parallel (redundant) network. Fiber-optic cables L1 and L2 on the FN2013 or FN2014 are interchangeable. L1 does not always have to be connected to L2. All fiber-optic cables in the diagrams below are connected with two fibers between the D23XX or FN2013/14 modules.

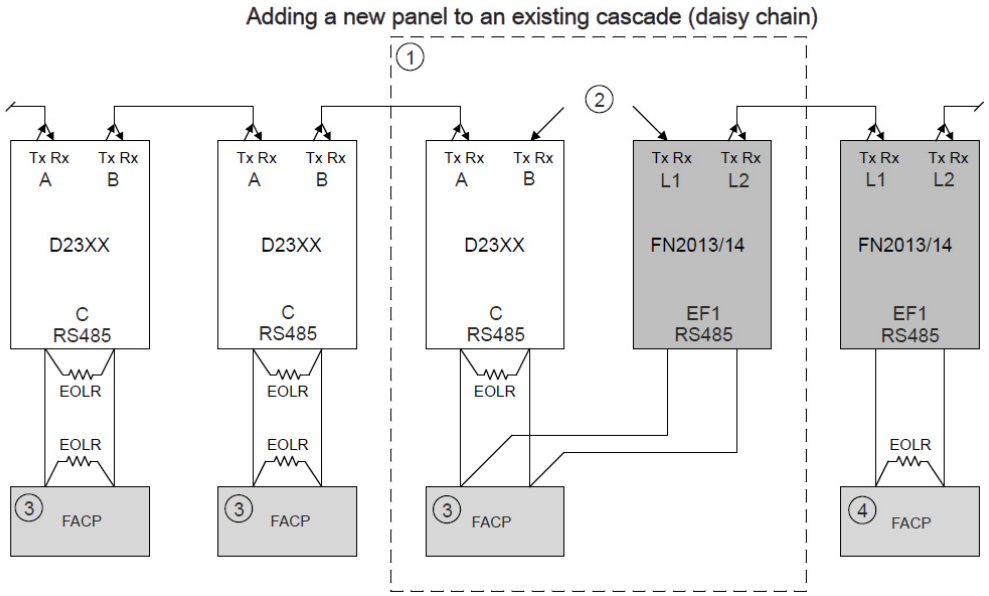


Fig. 15: Adding a new panel in cascade

L1 Fiber-optic SC connection for circuit 1 on FN2013/FN2014
L2 Fiber-optic SC connection for circuit 2 on FN2013/FN2014
EF1 Connection A1/B1, RS485 on FN2013/FN2014
A Fiber-optic SC connection circuit A on D23xx
B Fiber-optic SC connection circuit A on D23xx
C RS485 connection on D23xx
1 Shared housing
2 Fiber-optic cables are not permitted between D23xx and FN2013/FN2014
3 Existing fire control panel
4 New Fire control panel
EOLR Terminating resistor 120Ω

A terminating resistor is not necessary at the EF1 connection on FN2013/FN2014.

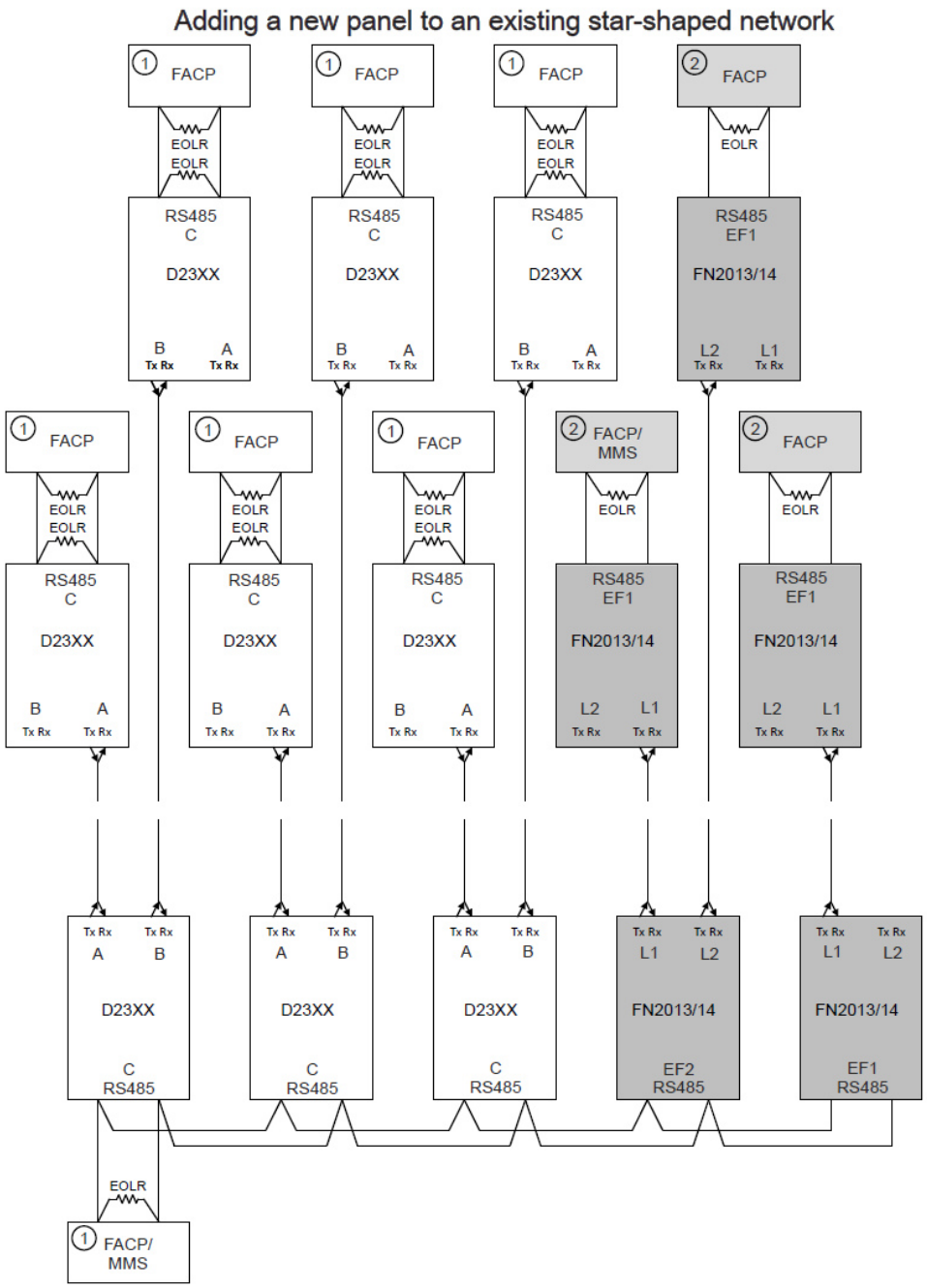


Fig. 16: New panel in existing star-shaped network

L1 Fiber-optic SC connection circuit 1 on FN2013/FN2014
L2 Fiber-optic SC connection circuit 2 on FN2013/FN2014
EF1 Connection A1/B1 RS485 on FN2013/FN2014
EF2 Connection A2/B2, RS485 on FN2013/FN2014
A Fiber-optic SC connection circuit on D23xx
B Fiber optic SC connection circuit B on D23xx
C RS485 connection on D23xx
1 Existing fire control panel or management station

2 New fire control panel or management station

EOLR Terminating resistor 120 Ω

FN2013 or FN2014 modules have an internal 120-ohm terminating resistor at the EF1 port. So that it can be used for a cascade (daisy chain connection) the EF2 port does not have this terminating resistor.

Connect only EF1 or EF2 on the same FN2013 or FN2014.

8 FCC Statement

Warning: Installation and usage of equipment is not in accordance with instructions manual:

- **Radiation of radio frequency energy**
- **Interference to radio communications**
  - **Install and use equipment in accordance with instructions manual.**
  - **Read the following information.**

This equipment generates, uses, and can radiate radio frequency energy and if not installed and used in accordance with the instructions manual, may cause interference to radio communications.

It has been tested and found to comply with the limits for a Class A computing device pursuant to Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference when operated in a commercial environment.

Operation of this equipment in a residential area is likely to cause interference in which case the user at his own expense will be required to take whatever measures may be required to correct the interference.

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